

Removing Reflexivity Alone Does Not Preserve Hammerstein Solvability

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Status. Candidate full counterexample, likely valid, pending expert review. The Hammerstein existence hypotheses (4.2)–(4.6) of Barroso–O’Regan do not remain sufficient when reflexivity is simply deleted. Failure occurs on real ℓ_1 with a constant kernel and a bounded globally Lipschitz nonlinearity, even when the growth ratio in (4.6) is zero.

1 Source question and precise answer

Barroso and O’Regan consider

$$y(t) = h(t) + \int_0^1 k(t, s) f(s, y(s)) ds, \quad 0 \leq t \leq 1, \quad (1)$$

where $f(s, \cdot)$ is weakly sequentially continuous, h is continuous, $k(t, \cdot) \in L^1$, and $t \mapsto k(t, \cdot)$ is L^1 -continuous. Their growth assumptions are

$$\|f(s, x)\| \leq \Omega(\|x\|), \quad K \limsup_{r \rightarrow \infty} \frac{\Omega(r)}{r} \leq 1, \quad K = \sup_t \int_0^1 |k(t, s)| ds. \quad (4.5\text{--}4.6)$$

They explain that the known proof for reflexive E uses weak compactness and ask what conditions are needed if reflexivity is removed [1, pp. 6–7].

space E this technique may fails due to lack of weak compactness in balls. Thus it

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is natural to ask what conditions are needed if the word reflexive is removed. Here as an application of our fixed point theory, we offer the following result as a partial answer to this question.

Figure 1: The question as printed across the bottom of page 6 and the top of page 7 of arXiv:math/0310422. Conditions (4.5)–(4.6) are transcribed above.

The question is open-ended. We settle its strongest natural yes/no interpretation negatively: conditions (4.2)–(4.6) alone do *not* imply solvability in an arbitrary Banach space. Some genuine replacement for reflexive weak compactness is necessary.

Proof intuition

With the constant kernel $k \equiv 1$ and $h = 0$, the right side of (1) does not depend on t . Every solution must therefore be constant, and the integral equation collapses to a fixed-point equation $c = F(c)$. We build a bounded fixed-point-free map F on ℓ_1 . Normally, norm continuity would not meet the required weak sequential continuity, but ℓ_1 has the Schur property: weak convergence of sequences already implies norm convergence.

2 A bounded fixed-point-free map on ℓ_1

Let B be the closed unit ball of the real space ℓ_1 . Define the radial retraction

$$R(x) = \frac{x}{\max\{1, \|x\|_1\}}, \quad x \in \ell_1,$$

and define $U : B \rightarrow \ell_1$ by

$$U(u) = (1 - \|u\|_1, u_1, u_2, u_3, \dots).$$

The first coordinate is nonnegative because $u \in B$.

Lemma 1. *The map $F = U \circ R : \ell_1 \rightarrow \ell_1$ has the following properties:*

1. $\|F(x)\|_1 = 1$ for every $x \in \ell_1$;
2. F has no fixed point;
3. F is globally 4-Lipschitz and weakly sequentially continuous.

Proof. For $u \in B$,

$$\|U(u)\|_1 = 1 - \|u\|_1 + \sum_{j=1}^{\infty} |u_j| = 1,$$

which proves the first assertion.

Suppose $U(u) = u$. Comparing coordinates gives

$$u_1 = 1 - \|u\|_1, \quad u_{j+1} = u_j \quad (j \geq 1).$$

The constant tail can lie in ℓ_1 only if $u_1 = 0$, so every coordinate is zero; the first equation then says $0 = 1$. Thus U has no fixed point. If $F(x) = x$, the first assertion gives $\|x\|_1 = 1$, hence $R(x) = x$ and $U(x) = x$, again impossible.

The elementary radial-projection estimate gives $\|R(x) - R(y)\|_1 \leq 2\|x - y\|_1$. Also

$$\|U(u) - U(v)\|_1 \leq |\|u\|_1 - \|v\|_1| + \|u - v\|_1 \leq 2\|u - v\|_1.$$

Therefore F is 4-Lipschitz. Finally, ℓ_1 has the Schur property: if $x_n \rightharpoonup x$, then $\|x_n - x\|_1 \rightarrow 0$ [2]. Lipschitz continuity now gives $F(x_n) \rightarrow F(x)$ in norm, and hence weakly. $\square$

3 The Hammerstein counterexample

Theorem 1. *There are data satisfying conditions (4.2)–(4.6) of the source on a nonreflexive Banach space for which (1) has no solution. One may take*

$$E = \ell_1, \quad h(t) = 0, \quad k(t, s) = 1, \quad f(s, x) = F(x),$$

where F is the map in Lemma 1. Moreover f is bounded and globally Lipschitz, and the growth factor in (4.6) equals zero.

Proof. Lemma 1 shows that $f(s, \cdot)$ is weakly sequentially continuous for every s . The zero function h is continuous. The constant kernel has $k(t, \cdot) = 1 \in L^1[0, 1]$, the map $t \mapsto k(t, \cdot)$ is constant in L^1 , and

$$K = \sup_{0 \leq t \leq 1} \int_0^1 |k(t, s)| ds = 1.$$

Since $\|F(x)\|_1 = 1$, choose the continuous nondecreasing majorant $\Omega(r) = 1$. Then

$$K \limsup_{r \rightarrow \infty} \frac{\Omega(r)}{r} = 0,$$

which is strictly stronger than the source’s bound.

Assume that $y \in C([0, 1], \ell_1)$ solves (1). The map $s \mapsto F(y(s))$ is norm continuous and therefore Bochner integrable. The equation reads

$$y(t) = \int_0^1 F(y(s)) ds.$$

Its right side is independent of t , so $y(t) = c$ for some $c \in \ell_1$. Substitution then gives

$$c = \int_0^1 F(c) ds = F(c),$$

contradicting Lemma 1. □

4 Scope, repair, and verification

Theorem 1 completely answers the removal-only formulation: even boundedness, global Lipschitz continuity, a constant kernel, and strict sublinear growth do not replace reflexivity. The obstruction is exactly the loss of weak compactness needed by a fixed-point argument.

This does not characterize all sufficient replacement conditions, and it does not contradict Theorem 4.1 of the source. That theorem imposes an additional ϕ -contractive/condensing hypothesis; our F is deliberately fixed-point free and does not satisfy it. The counterexample instead proves that an extra condition of this general kind is indispensable.

The proof is noncomputational. Review reduces to four explicit checks: the fixed-point-free coordinate recurrence, the Schur-property continuity step, the five source hypotheses, and the constant-solution reduction. Cheap local indexes and bounded primary-source searches through August 11, 2026 found no explicit answer to the source prompt by this construction. Since the shift map on the ℓ_1 ball is classical, novelty is asserted only for its Hammerstein encoding and remains subject to expert review.

References

- [1] C. S. Barroso and D. O'Regan, *Measures of Weak Compactness and Fixed Point Theory*, arXiv:math/0310422v2 (2003), Section 4.
- [2] I. Schur, "*Über lineare Transformationen in der Theorie der unendlichen Reihen*", J. Reine Angew. Math. **151** (1921), 79–111.